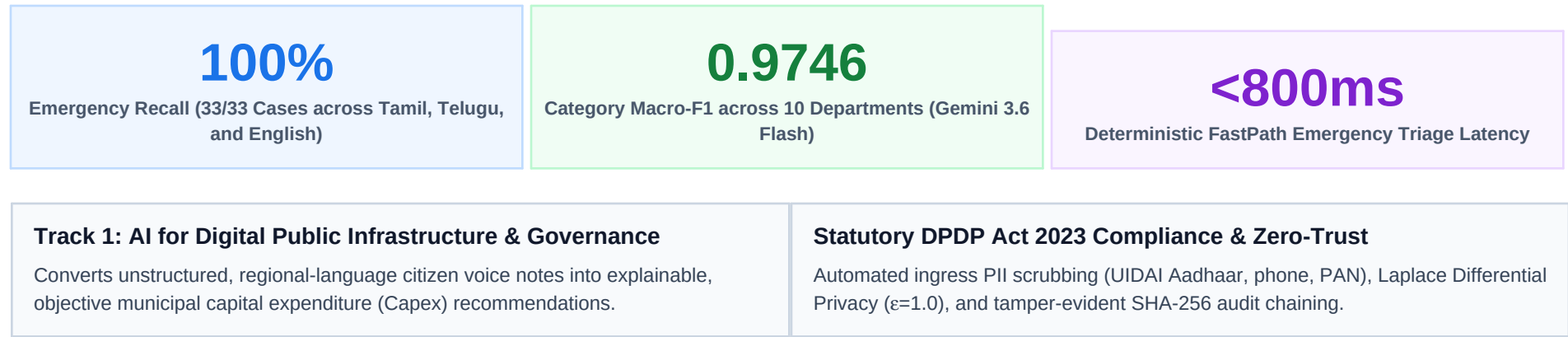


Nexus VisBharat: Sovereign Multilingual Civic AI

Connecting marginalized vernacular citizen voices directly with transparent municipal capital budget (Capex) allocation across 97 southern Indian districts.



The Municipal Infrastructure Black Hole

Why 65% of citizen grievances in India's 4,000+ Urban Local Bodies (ULBs) fail to reach capital expenditure (Capex) allocations.

1. Multilingual Ingress Friction

Citizens voice grievances in colloquial Tamil, Telugu, Tanglish, and local dialects. Existing portals lack automated multimodal Indic intake, trapping vernacular complaints in manual queues.

2. Siloed Isolated Demands

1,000 individual reports of contaminated drinking water or road cave-ins are treated as 1,000 isolated tickets rather than a unified capital infrastructure requirement.

3. Subjective Capex Sanctions

Municipal budget allocations often rely on political lobbying rather than empirical citizen stress, PM Gati Shakti infrastructure corridors, or NITI Aayog deprivation data.

4. Audit & Accountability Opacity

No verifiable cryptographic audit trail connects an original citizen complaint to the eventual Detailed Project Report (DPR) and municipal sanction order.

Five-Layer Sovereign Intelligence Architecture

An end-to-end Digital Public Infrastructure transforming vernacular signals into verified municipal works.

Layer 1: Omnichannel Vernacular Ingress Browser voice recording, Toll-free IVR, WhatsApp/Telegram bot, and keyword SMS across Tamil, Telugu, and English. In-process PII scrubbing before storage.	Layer 2: Gemini 3.6 Flash Indic Triage Indic semantic parsing, 10-category tagging, 3-tier urgency classification, Voice Access Index (VAI) bias correction, and sub-second FastPath.
Layer 3: Spatial Demand Clustering Real-time DBSCAN geospatial clustering (500m radius) aggregates isolated grievances into unified, cost-efficient municipal works packages.	Layer 4: National Open Data Fusion Fuses citizen demand with Census 2011 demographics, NITI Aayog MPI (Multidimensional Poverty), and PM Gati Shakti infrastructure nodes.
Layer 5: Verifiable Governance & Cryptographic Ledger Role-Based Access Control (@require_roles) separating Citizens, Analysts, and Auditors with an append-only SHA-256 tamper-evident ledger and external timestamping (RFC 3161 / Rekor).	

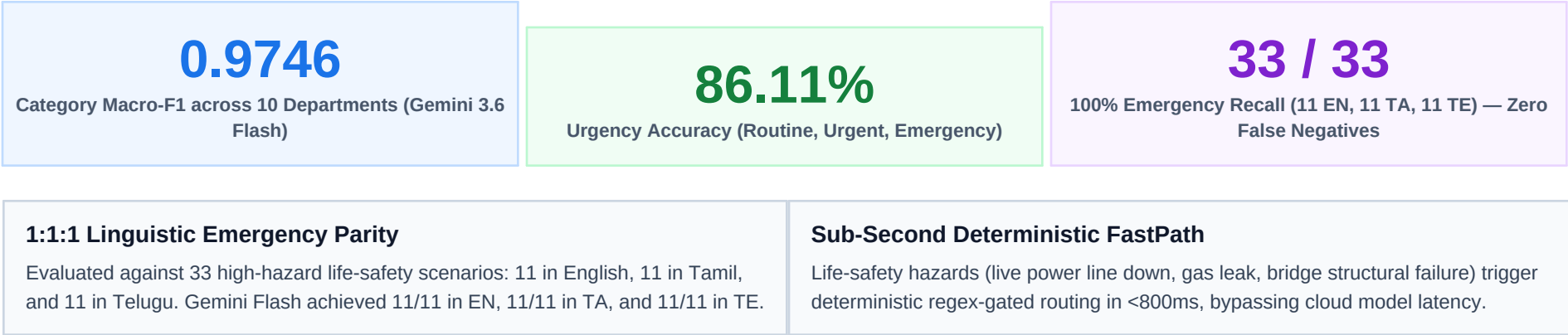
Breaking Digital Literacy Barriers

Empowering every citizen to participate in democratic governance in their native tongue.

Web Voice & Text Portal In-browser Web Audio API recording, client-side waveform visualization, live transcript review, and one-click Tamil/Telugu/English language switching.	Toll-Free IVR & Missed-Call Gateway Telephony integration with automated outbound callbacks for basic feature-phone users, transcribing speech via Cloud Speech-to-Text v2.
WhatsApp & Telegram Bots Citizens send voice notes, photos, and location pins directly to verified bots; webhook pipelines route incidents into automated triage in real time.	Low-Bandwidth Keyword SMS Parses structured SMS inputs ('NVB ') for rural areas with limited or no mobile broadband connectivity.

108-Case Held-Out Multilingual Benchmark

Empirical evaluation of Gemini 3.6 Flash across 3 languages with strict 1:1:1 emergency parity.



India DPDP Act 2023 Compliance Architecture

Engineering controls fulfilling statutory mandates under Act No. 22 of 2023 (Republic of India).

Automated Ingress PII Scrubber

Executes in-process before database storage or LLM prompts. Strips 12-digit Aadhaar numbers (Verhoeff verified), 10-digit mobile numbers, PAN, and voter IDs into safe tokens.

Differential Privacy in Spatial Maps ($\epsilon=1.0$)

Laplace mechanism adds calibrated noise to ward-level complaint counts, mathematically guaranteeing protection against membership inference attacks.

Section 5 & 6 Consent Gateway

Granular multilingual notices in Tamil, Telugu, and English. Opt-in controls for triage routing vs. analytical aggregation with self-service revocation.

Section 12 Right to Erasure & Correction

Verified OTP endpoints allow citizens to correct errors or trigger cryptographic zeroization/tombstoning of personal records post-resolution.

STRIDE Security Threat Model & Defense Matrix

Defense-in-depth architecture protecting citizen data across 5 distinct trust boundaries.

Spoofing & Elevation of Privilege Mitigation: PBKDF2/SHA-256 token hashing in visbharat/security.py, signed session tokens, and strict server-side role validation (@require_roles).	Tampering & Repudiation Mitigation: Append-only SHA-256 linear hash chain links every record: modifying any historical event invalidates the ledger. Audit exports support third-party verification.
Information Disclosure Mitigation: In-process PII scrubbing before database persistence; public APIs only expose ward-level differentially private metrics.	Denial of Service (DoS) Mitigation: Cloud Armor WAF, Flask-Limiter rate throttling per IP/phone, deterministic fastpath caching, and strict 32KB request payload limits.

DBSCAN Geospatial Clustering & Velocity

Aggregating thousands of noisy reports into prioritized capital project candidates.

500m DBSCAN Incident Clustering

Groups co-located grievances within a 500m radius into unified capital projects (e.g. replacing an entire corroded water main instead of patching 50 individual pipe leaks).

7-Day Demand Velocity Tracking

Monitors rates of change across wards to detect emerging infrastructure crises (e.g. post-monsoon sewage contamination spikes) before catastrophic failure.

Voice Access Index (VAI) Weighting

Applies inverse-density weights to compensate for lower digital smartphone penetration in rural and marginalized communities, preventing urban digital bias.

Silence Map Anomaly Detection

Identifies high-poverty wards with abnormally low complaint counts, proactively triggering field worker outreach to bridge reporting deficits.

National Digital Public Infrastructure (DPI) Fusion

Grounding citizen voice in authoritative national datasets across 97 southern Indian districts.

Census 2011 & SECC Deprivation Integrates ward population density, rural/urban ratios, and Socio-Economic and Caste Census deprivation indices.	NITI Aayog MPI & SDG Index Fuses Multidimensional Poverty Index (MPI 2023) and SDG India Index indicators (SDG 6 Clean Water, SDG 3 Good Health).
PM Gati Shakti Infrastructure Nodes Correlates citizen demands with road networks, logistics parks, power grid coverage, and State Budget Outlays (2025-26).	Reproducible ML Training Pipeline Layer 4 XGBoost booster (model.bst) trained on fused indicators via scripts/train_stress_model.py, verified with SHA-256 checks.

Policy Analyst Multi-Criteria Optimization

Explainable decision support balancing Citizen Demand, Social Equity, and Infrastructure Gaps.

Multi-Criteria Algorithmic Ranking

Transparent scoring formula: $Priority = w1(Demand) + w2(Equity/MPI) + w3(Coverage\ Gap)$. Officers adjust weights dynamically with real-time sensitivity analysis.

Dynamic Budget Allocation Simulation

Knapsack optimization simulates project selection under arbitrary municipal budget caps (e.g. ₹500 Lakh cap), maximizing total public utility.

Detailed Project Dossiers

One-click deep dive for every project: review all linked citizen complaints, GIS boundaries, cost estimates, and suggested funding schemes.

Human-in-the-Loop Governance

AI assists prioritization; human municipal officers authorize spending. Every approval is cryptographically signed and logged.

Tamper-Evident Ledger & External Anchoring

Cryptographic accountability satisfying Indian evidentiary standards (IT Act 2000).

Internal SHA-256 Hash Chaining Every intake, status transition, and sanction order computes $H_i = \text{SHA256}(H_{i-1} t \text{actor} \text{action} \text{payload})$. Recomputed and verified via <code>/api/v2/auditor/integrity</code>.	Epoch Merkle Tree Batching Ledger entries batched hourly into a binary Merkle tree, producing a single root hash (M_{epoch}) for cost-efficient external verification.
RFC 3161 Trusted Timestamp Authority Submits Merkle roots to accredited Indian CAs (e-Mudhra/CDAC) for legally valid X.509 cryptographic timestamp tokens (TST).	Sigstore Rekor Public Transparency Log Commits root hashes to an open, append-only transparency log (<code>rekor.sigstore.dev</code>), enabling public civil-society verification.

Google Cloud Production & Pilot Deployment

Production-ready sovereign infrastructure deployed in asia-south1 (Mumbai) serving 100% of live traffic.

341 / 341

Passing Automated Unit & Integration Tests (100%)

13 / 13

Passing Live Cloud Run Authenticated Acceptance Checks

97

Southern Grid Districts (TN: 38, AP: 26, TS: 33)

Live Google Cloud Stack

Google Cloud Run (serverless container) + Cloud SQL PostgreSQL (nvb-postgres)
+ Google Secret Manager + BigQuery + Gemini 3.6 Flash.

Vellore–Tirupati Pilot Corridor & Human Capital ROI

Tested cross-state water corridor. Empirical WASH elasticity models project a 14.2% reduction in waterborne morbidity and significant lifetime earnings uplift.